

Adaptive cooperative detection method for unmanned planetary vehicles based on deep reinforcement learning

Peng Yan
Department of Aerospace Engineering
Harbin Institute of Technology
Harbin, China
yanpeng@hit.edu.cn

Hewen Xiao
Department of Control Science and
Engineering
Harbin Institute of Technology
Harbin, China
p_xiaohewen@163.com

Jifeng Guo
Department of Aerospace Engineering
Harbin Institute of Technology
Harbin, China
guojifeng@hit.edu.cn

Chengchao Bai
Department of Aerospace Engineering
Harbin Institute of Technology
Harbin, China
baichengchao@hit.edu.cn

Hongxing Zheng
Department of Aerospace Engineering
Harbin Institute of Technology
Harbin, China
Zhenghongxing_hit@aliyun.com

Abstract—Unmanned planetary vehicles have important significance for human development and utilization of space resources through the detection of extraterrestrial objects. This paper proposes an adaptive cooperative detection method for unmanned planetary vehicles, which can be adapted to the complex and changeable planetary environment through self-learning methods. First, the detection environment is rasterized; then, the convolutional neural network is used to process the detection area information and the lidar information to extract the environmental features; the reinforcement learning method is used to learn the strategy of detecting the environment. The reward function was designed in detail so that the planetary vehicles can quickly detect a given area without collision. At the same time, the detection information can be shared among planetary vehicles to accelerate the learning process. Finally, simulation experiments were carried out in Gazebo. The experimental results show that the planetary vehicles can detect the given area quickly and effectively.

Keywords—planetary vehicles, cooperative detection method, self-learning, deep reinforcement learning

I. INTRODUCTION

With the advancement of science and technology, human exploration of unknown space has gradually increased. At present, a large number of interstellar detectors have been launched to explore the planets in the solar system. Planetary vehicles have important scientific significance for human to study the evolution of planetary life, develop and utilize space resources by directly detecting extraterrestrial objects, exploring climate and geological characteristics and searching for life signals [1].

In the process of exploring complex and unknown planetary environments, planetary vehicles need the ability of autonomous perception of planetary environments and path planning to ensure the success of missions. As a key technology of autonomous capability of planetary vehicles, autonomous navigation method of planetary vehicles has been widely studied [2-5]. On the basis of a large number of studies, a single planetary vehicle can perform some simple tasks. However, due to the complexity of the planetary environment, a single planetary vehicle can perform limited tasks. In extreme cases, it is likely to fail due to unexpected failures. Multi-planetary vehicles can share detection information through collaborative work, which can rapidly enhance the awareness of the unknown unstructured

planetary environment, expand detection speed of planetary vehicles, and improve the success rate of missions.

Multi-robot collaboration methods have been extensively studied [6-8]. In [9], a cooperative method considering the limitation of communication was studied for area exploration. The problem of global exploration in unknown environment is divided into multiple trajectory planning problems in local known environment. The robot's local destination is the optimal boundary point, which maximizes information gain, minimizes navigation costs, and satisfies communication constraints. In [10], a decentralized cooperation control method for connectivity maintenance of the mobile robots communication network was studied. The authors proposed a gradient-based control method that utilizes decentralized estimation of the algebraic connectivity, which solves the global connectivity problem in a decentralized manner. In [11], the problem of formation control of multi-robot based on the virtual structure method and behavioral approach was studied. In the virtual structure, the mobile robot is modeled by an electric charge. Each mobile robot can find its location in the formation autonomously. The behavioral structure is designed to avoid obstacles. There is a common disadvantage of the above methods. They do not have the self-learning ability, and can't dynamically adjust their strategies in unknown environments, so they are not suitable for planetary exploration.

Reinforcement learning [12] has a wide range of applications due to its powerful self-learning ability [13,14]. Deep reinforcement learning is a combination of deep learning and reinforcement learning. It utilizes the powerful feature extraction ability of deep learning and the self-learning ability of reinforcement learning, and it has close or exceed human performance in many fields [15,16]. In recent years, deep reinforcement learning has been extensively studied in robot applications. In [17-19], the authors did a series of studies on the navigation problems of mobile robots in pedestrians. The authors used a deep reinforcement learning method to train the network model offline and used the trained model to navigate online. Experimental results show that trained models can navigate autonomously in pedestrians. In [20], a multi-robot collision avoidance method based on raw lidar data was proposed. The authors used a strategy-gradient-based reinforcement learning algorithm to map sensor data directly into control commands

via deep neural networks. The simulation results in different scenarios verify the effectiveness of the algorithm.

In addition, deep reinforcement learning has a lot of research in the field of multi-robots. In [21], the authors proposed a multi-agent asynchronous advantage actor-critic algorithm, which is a method of centralized training, distributed execution in a fully observable environment, and sharing the experience of all agents to speed up training. The authors verified the applicability of the algorithm on the multi-agent collaborative construction task. In [22], The authors tested the performance of several deep reinforcement learning methods to learn multi-agent cooperative strategies in a variety of complex and partially observable environments. During the testing process, the effects of network structure, training process and reward function structure on learning performance were compared and analyzed. In [23], the authors studied how to create a cooperative and competitive relationship between agents using deep reinforcement learning methods by setting reward functions.

The main contributions of this work are (i) an adaptive cooperative detection method based on deep reinforcement learning which enables planetary vehicles to have the ability of self-learning and adjust their strategies according to different environments, (ii) a detection experience sharing method which accelerates the learning process by sharing information among planetary vehicles.

II. PROBLEM MODEL

The problem of two planetary vehicles cooperatively detect unknown area is studied. Given an unknown area E and two planetary vehicles $c_i (i = 1, 2)$, the problem is how to detect the unknown area in the shortest time t . The planetary vehicles are equipped with a lidar and a visual camera to detect the given area. There are uncertain obstacles $O_i (i = 1, 2, \dots, n)$ in the area. The planetary vehicles perceive obstacles through the lidar to avoid collision with obstacles and detect unknown area by taking pictures with the visual camera. At the same time, in order to maintain communication between the two planetary vehicles, the distance d between them must be within a certain communication range $d_{\max}$. The detection range of a planetary vehicle can be simplified to a sector, represented by the detection radius $r_{\det}$ and angle $\theta_{\det}$. The problem can be expressed by the following formula:

$$\begin{aligned} \min \quad & t \\ \text{s.t.} \quad & (E_1^1 \cup E_i^2) \geq E \\ & P_i \notin O_j (i = 1, 2; j = 1, 2, \dots, n) \\ & d < d_{\max} \end{aligned} \quad (1)$$

where E_i^1 and E_i^2 are the areas that the two planetary vehicles have detected at time t , respectively. P_i is the position of planetary vehicle i .

The problem of two planetary vehicles cooperatively detect unknown area is shown in Fig. 1.

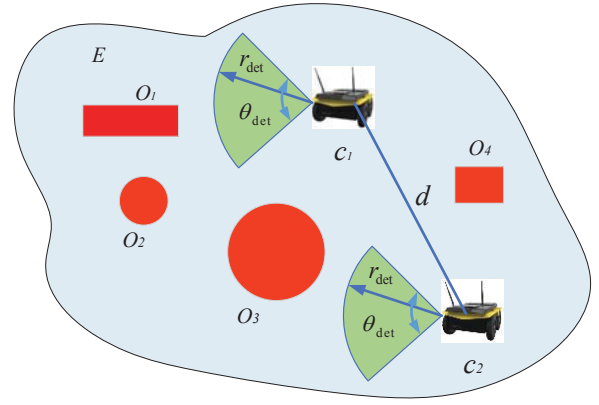


Fig. 1. Two planetary vehicles detect unknown area.

As shown in Fig. 1, the blue area is the area to be detected, the red areas are randomly distributed obstacles, and the green sectors are the detection capability range of the planetary vehicles.

III. ADAPTIVE COOPERATIVE DETECTION METHOD

Deep Q-Network(DQN) [15] is used to enable the planetary vehicles have the ability of detecting unknown area cooperatively. DQN combines the self-learning ability of the reinforcement learning with the feature extraction ability of the deep neural network. It has the ability to compete with humans on many tasks. Compared with other deep reinforcement learning algorithms, DQN can learn online and adjust its strategy according to the environment in real time, which is suitable for the task of planetary vehicles detecting unknown environment.

A. Representation of Detection Area

The detection area is rasterized as shown in Fig. 2.

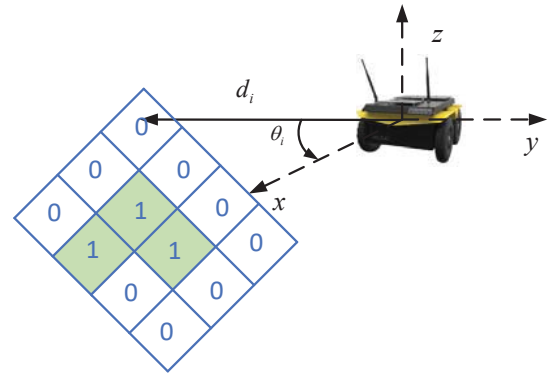


Fig. 2. Representation of detection area.

As shown in Fig. 2, each grid i has three kinds of information: d_i , which represents the distance of the grid i from the planetary vehicle; θ_i , which represents the angle between the line connecting the grid i and the center of mass of the planetary vehicle and the forward direction of the planetary vehicle; $flag_i$, which represents whether the grid i has been detected. If $flag_i = 1$, the grid i has been detected, and if $flag_i = 0$, the grid i has not been detected. Assuming that the detection area is rasterized into K grids, the detection coverage rate η can be expressed as

$$\eta = \frac{\sum_{i=1}^K \text{flag}_i}{K} \quad (2)$$

B. DQN Algorithm

In reinforcement learning tasks, agents learn through interaction with the environment. The agent observes the environment state S_t , takes action A_t , and then changes the environment state to S_{t+1} . At the same time, the agent obtains reward R [10]. The goal of the agent is to select actions which maximize cumulative future reward. We use a deep neural network to approximate the optimal action-value function

$$Q^*(s, a) = \max_{\pi} \mathbb{E} [r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots | s_t = s, a_t = a, \pi] \quad (3)$$

where $\pi = P(a | s)$ is behaviour policy, γ is discount rate.

During detecting the area, the experiences information $e_t = (s_t, a_t, r_t, s_{t+1})$ of the two planetary vehicles are stored in the same data set $D_t = \{e_1, \dots, e_t\}$ through communication. In the process of learning, the network is updated by Q-learning method with minibatches of experience $(s, a, r, s') \sim U(D)$.

The network loss function is:

$$L_i(\theta_i) = \mathbb{E}_{(s, a, r, s') \sim U(D)} \left[\left(r + \gamma \max_{a'} Q(s', a'; \theta_i^-) - Q(s, a; \theta_i) \right)^2 \right] \quad (4)$$

where θ_i are the parameters of the Q-network at iteration i and θ_i^- are the target network parameters at iteration i . The target network parameters θ_i^- are only updated with the Q-network parameters every C steps and are held fixed between individual updates.

C. Action Space

The command action of the planetary vehicles consists of an angular velocity command and a linear velocity command. The angular velocity command is discretized as $(-0.5, 0, 0.5)$ rad/s and the linear velocity command is discretized as $(0, 0.5)$ m/s. The command action space of planetary vehicles is composed of the angular velocity command and the linear velocity command. There are a total of six command actions. Fig. 3 shows the linear velocity command v_c and angular velocity command ω_c of the planetary vehicle.

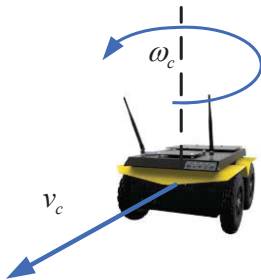


Fig. 3. Planetary vehicle action command.

D. Network Architecture

When learning with DQN, the deep neural network needs to process the information of the lidar, the information of detection area and the motion information of planetary vehicles. The information of the lidar includes three consecutive scans, each of which contains 180 data. The scanning angle ranges from -90° to 90° and the scanning distance ranges from 0.1m to 10.0m. The information of the detection area contains three matrices, which respectively indicate the distance matrices $\mathbf{D}$, angle matrices $\mathbf{\Theta}$ and flag matrices $\mathbf{F}$, where $\mathbf{D} = [d_1, d_2, \dots, d_{100}]$, $\mathbf{\Theta} = [\theta_1, \theta_2, \dots, \theta_{100}]$, $\mathbf{F} = [\text{flag}_1, \text{flag}_2, \dots, \text{flag}_{100}]$. We use subscript 1 for the current planetary vehicle and subscript 2 for the other planetary vehicle. Taking planetary vehicle 1 as an example, the motion information of planetary vehicle 1 includes the linear and angular velocities of planetary vehicle 1, the linear and angular velocities of planetary vehicle 2, and the position of planetary vehicle 2 in the system (as shown in Fig. 2) of planetary vehicle 1, which is expressed by polar coordinates as distance d_1^2 and angle ϕ_1^2 . Convolutional neural network (CNN) [24] is used to process the lidar information and the detection area information. The network architecture is shown in Fig. 4.

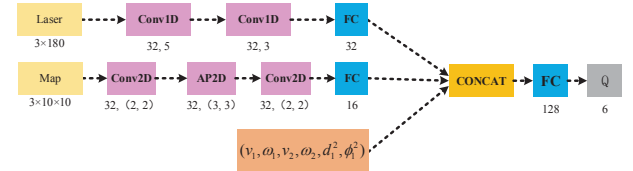


Fig. 4. Network architecture.

As shown in Fig. 4, three hidden layers are used to process the lidar information. The first hidden layer has 32 one-dimensional filters with kernel size = 5, stride = 2. The second hidden layer has 32 one-dimensional filters with kernel size = 3, stride = 2. The third hidden layer is a fully connected layer with 32 nonlinear activation functions. The information of detection area are processed by four hidden layers. The first hidden layer has 32 two-dimensional filters with kernel size = (2, 2), stride = 1. The second hidden layer is a two-dimensional average pooling layer with kernel size = (3, 3), stride = 1. The third hidden layer is the same as the first hidden layer. The fourth hidden layer is a fully connected layer with 16 nonlinear activation functions. After the above processing, the information of the lidar and the detection area are concatenated with the motion information of planetary vehicles. A full-connection layer with 128 nonlinear activation functions and a full-connection layer with 6 linear activation functions are used to obtain the Q value of each action. The nonlinear activation function used in the above network is ReLU [25].

E. Reward Function

Our goal is to enable two planetary vehicles to detect a given area at the shortest time and avoid obstacles in the detection area during the detection process. The two planetary vehicles need to be kept within a certain range, so that planetary vehicles can't collide with each other and can maintain normal communication. Based on the above considerations, the reward function is designed as follows:

$$r_i^t = (r^g)_i^t + (r^c)_i^t + (r^d)_i^t \quad (5)$$

where $(r^g)_i^t$, $(r^c)_i^t$ and $(r^d)_i^t$ are area detection reward, collision avoidance reward and communication maintenance reward received by the planetary vehicle i at time t , respectively.

The area detection reward is:

$$(r^g)_i^t = \begin{cases} 10(\eta_t - \eta_{t-1}) & \text{if } \eta_t - \eta_{t-1} > 0 \\ -0.05 & \text{if } \eta_t - \eta_{t-1} = 0 \\ 10 & \text{if } \eta_t > 0.85 \end{cases} \quad (6)$$

This equation indicates that when the detection coverage rate increases, the planetary vehicle will receive a positive reward, when the detection coverage rate remains unchanged, a negative reward, and when the detection coverage rate exceeds 0.85, a large positive reward, indicating that the task has been completed.

The collision avoidance reward is:

$$(r^c)_i^t = \begin{cases} -2 & \text{if } \|P_1 - P_2\| < d_{\min} \text{ or } \|P_i - O_j\| < d_{\min} \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

where $d_{\min}$ is minimum safety distance. This equation indicates that when the distance between the planetary vehicle is less than the minimum safety distance or the distance between the planetary vehicle and the obstacle is less than the minimum safety distance, the planetary vehicle will receive a negative reward.

The communication maintenance reward is:

$$(r^d)_i^t = -0.1 \quad \text{if } d > d_{\max} \quad (8)$$

where $d_{\max}$ is the maximum communication distance between planetary vehicles. This equation indicates that when the distance between planetary vehicles exceeds the maximum distance of communication, the planetary vehicle will receive a negative reward.

IV. SIMULATION RESULTS AND ANALYSIS

A. Simulation Environment and Training Parameters

A simulation environment was built for the planetary vehicles in the Gazebo for network training and algorithm testing, as shown in Fig. 5.

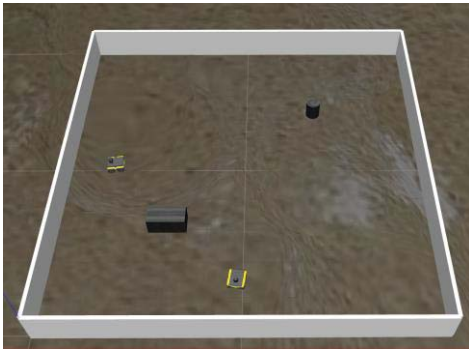


Fig. 5. Simulation environment.

As shown in Fig. 5, the area that the planetary vehicles need to detect is a rectangular area of 10m×10m in which obstacles are randomly distributed. The planetary vehicles perceive the obstacles through the lidar and use visual cameras to obtain area information.

We trained deep neural networks in the above simulation environment. The training parameters are listed in the TABLE 1.

TABLE I. TRAINING PARAMETERS.

Parameters	Values
Maximum communication distance $d_{\max}$	5m
Detection radius r_{det}	3m
Detection angle θ_{det}	60deg
Total number of grids K	100
Target network update period C	500steps

B. Results and Analysis

The training process took 2,000 training episodes and lasted 4.6 hours. The position of the obstacle and the planetary vehicles are randomly reset before the start of each training episode. The maximum number of training steps in one episode is 500 steps. The stopping condition of a training episode is that the maximum number of training steps is reached or the detection coverage rate exceeds 0.85.

We performed 10 random tests using the trained deep neural network. The results are shown in the Fig. 6 :

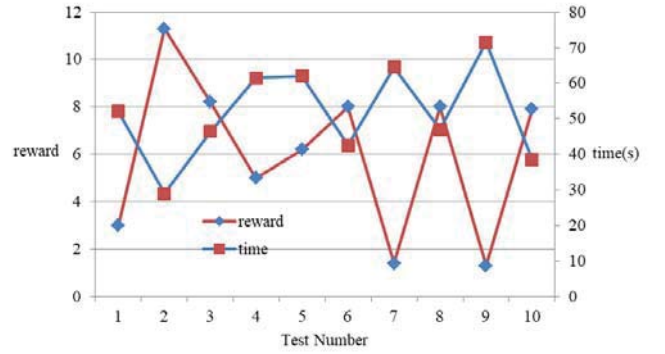


Fig. 6. 10 random test results.

Fig. 6 shows the results of 10 random tests, in which the horizontal axis represents the number of the tests, the vertical axis represents the accumulated reward value and detection time during a test. In the above 10 random tests, each detection coverage is 0.86, indicating that the planetary vehicles can complete the detection task. As shown in Fig. 6, since the initial positions of the planetary vehicles and the positions of obstacles are randomly set, the detection time varies with the environment. At the same time, it can be seen from the trend of the curve that the longer the detection time, the smaller the cumulative reward value, which is consistent with the reward function.

It is worth noting that the cumulative reward value of the seventh and ninth tests is significantly smaller than the cumulative reward value of the other tests, but the detection time is not much different from the other tests. There are two reasons for this phenomenon. First, during the test, the two planetary vehicles were not within communication range for a while, so they received negative rewards during the process. Second, the detection areas of the two planetary vehicles have overlapping parts during the detection process. This phenomenon can be seen in Fig. 7.

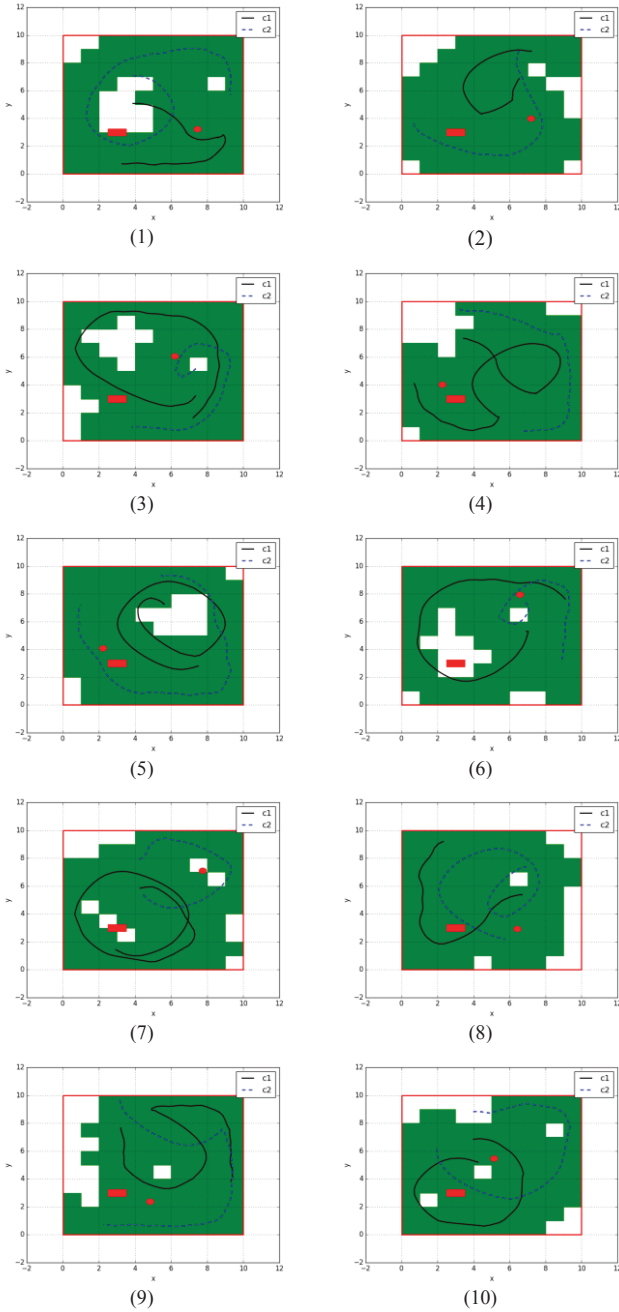


Fig. 7. The process of 10 random area detections.

Fig. 7 shows the process of 10 random area detections. The green areas are the detected areas, the red areas are the obstacles, and the blue and the black solid lines are the trajectories of the planetary vehicles. As shown in Fig. 7, in most cases, the areas detected by the two planetary vehicles do not intersect very much, and most of the areas are independently detected. This result reflects the cooperative behavior of planetary vehicles in the detection process, which can increase the efficiency of area detection. In the seventh and ninth detection processes, the detection areas of the two planetary vehicles intersect greatly, which increases the detection time and reduces the cumulative rewards. At the same time, in the seventh and ninth tests, the cumulative rewards for the two tests were reduced because the two planetary vehicles were out of communication range for a period of time.

V. CONCLUSION

In this paper, an adaptive cooperative detection method based on deep reinforcement learning is proposed to detect unknown areas for planetary vehicles. The planetary vehicles are equipped with a lidar and a visual camera to detect an unknown area. The lidar perceives obstacles in the area to achieve obstacles avoidance; the visual camera collects pictures to achieve area detection. The detection area is rasterized to facilitate the representation of the detection area and the calculation of the detection coverage rate. In order to extract environmental features, the CNN network is used to process the lidar information and the detection area map information. By designing the reward function, the planetary vehicles can learn a strategy to detect a given area quickly and without collision. Finally, the simulation environment was built in Gazebo, and the deep neural network was trained and tested in this simulation environment. The test results show that the algorithm can make the planetary vehicles cooperatively detect the given area and avoid obstacles. Future work will extend this method to dynamic environment and make a fusion of information from the lidar and visual camera to avoid obstacles.

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